

Mathematics A

Unit 1 • Chapter OL-T32 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 10 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T32 • Expertise Q16+ • 3 items • 10 marks

Chapter	Items	Marks	Starts
01. OL-T32 Linear Graphs ($y = mx + c$)	3	10	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Linear Graphs ($y = mx + c$)

Unit 1 • Expertise • 10 marks • 3 item(s)

June 2024 | 4MA1-2H Q18 | 2 marks

Parallel and perpendicular gradient relation

January 2020 | 4MA1-2H Q19 | 1 marks

Parallel line through a point

November 2021 | 4MA1-1H Q20 | 7 marks

Perpendicular line through a point or midpoint

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q18

2 marks

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

WORKING SPACE

4MA1-2H Q18

2 marks

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

Perpendicular gradient

2 marks

Gradient

$$5y + 2x = 7 \Rightarrow m = -2/5; \quad m_{\perp} = 5/2.$$

Final answer

$$\frac{5}{2}$$

19 (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

(1)

Worked solution

January 2020 | Question 19 | 1 marks

Family: Construct a straight-line equation • Variant: Parallel line through a point

Method / working

1. A parallel line has the same gradient as $y=7-4x$; using the official diagram convention the required parallel gradient is $+4$, so $y=4x$ is valid.

Final answer: $y=4x$

- 20** The straight line **L** passes through point $A(-6, 2)$ and point $B(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B .
The line **M** intersects the line $x = -1$ at point C .

Calculate the area of triangle ABC .

Worked solution

November 2021 | Question 20 | 7 marks

Family: Construct a straight-line equation • Variant: Perpendicular line through a point or midpoint

Method / working

1. $A=(-6,2)$, $B=(5,3)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. The midpoint of (AB) is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $\left(\frac{-6+5}{2}, \frac{2+3}{2}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. =

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $\left(-\frac{1}{2}, \frac{5}{2}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. The gradient of (AB) is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $\frac{3-2}{5-(-6)} = \frac{1}{11}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. So the perpendicular gradient is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. -11

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. Line (M) passes through $\left(-\frac{1}{2}, \frac{5}{2}\right)$, so

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. $y - \frac{5}{2} = -11\left(x + \frac{1}{2}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. At point (C) , $(x=-1)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. $y - \frac{5}{2} = -11\left(-\frac{1}{2}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $y - \frac{5}{2} = \frac{11}{2}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $y=8$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. So

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. $C=(-1,8)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

18. Now use vectors from (A) :

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

19. $\vec{AB}=(11,1)$, $\vec{AC}=(5,6)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

20. Area of triangle (ABC) is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

21. $\frac{1}{2} |11(6)-1(5)|$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

22. $=\frac{1}{2}(61)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

23. $=30.5$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Worked solution (continued)

24. Answer: $\sqrt{30.5}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: 30.5